

This query sorts all students in ascending order by age. The query runs successfully because `age.keyword` is a `keyword` type field on which we can apply a sorting function. By default, we sort on the second data type (`age.keyword`) created by Elasticsearch.

> **NOTE** In the example, `age.keyword` is the default name provided by Elasticsearch during dynamic mapping which is derived implicitly. Whereas, we have full control over creating fields, their names, and their types when defining mapping schemas explicitly.

Treating a field as a `text` type instead of a `keyword` unnecessarily affects the engine as data is analyzed and broken down into tokens.

Deriving incorrect date formats

Another potential problem with dynamic mapping is that date formats may be determined incorrectly if not provided in Elasticsearch's default date format (`yyyy-MM-dd` or `yyyy/MM/dd`). A date is considered a `text` type if we post it in the United Kingdom's `dd-MM-yyyy` format or the United States' `MM-dd-yyyy` format. We cannot perform date math on fields associated with non-date data types. Such fields are also ineligible for sorting, filtering, and aggregations.

> **NOTE** There is no `date` type in JSON, so it's up to consuming applications to decode the value and determine whether it is a date. When working with Elasticsearch, we provide data in a string format for these fields: for example, `"release_date" : "2021-07-28"`.

The takeaway when working with dynamic mapping features is that there is a scope for erroneous mapping by Elasticsearch. Preparing our schema based on field values may not always fit the bill. So, the general advice is to develop the schema based on our data model requirements rather than relying on the mercy of the engine.

To overcome those limitations, we can choose an alternative: *explicit mapping*, where we define our schema and create it before the indexing process kicks in. We discuss explicit mapping in the next section.

4.3 Explicit mapping

In the previous section, we discussed Elasticsearch's schema-less dynamic mapping. Elasticsearch is intelligent enough to derive mapping information based on our documents, but we can end up with an incorrect schema definition. Fortunately, Elasticsearch provides ways for us to dictate the mapping definitions.